AI Log - Moneyness, Decision Time, and Metric Interpretation

Representative conversation: clarifying binary-contract moneyness, decision-time meaning, and a likely plot-logic issue.

Context

After the dashboard was mostly working, the user asked conceptual questions about how to interpret decision time and moneyness in the context of short-dated binary BTC contracts.

Decision-time clarification

The user asked whether average decision minutes meant time between open and decision, or time between decision and expiry. The assistant clarified that the metric came from minutes_before_close_at_decision, so it measures how many minutes before close the contract became effectively decided.

Binary moneyness explanation

The assistant explained that for a binary contract, moneyness is still essentially underlying/spot/futures price versus the threshold or strike. The difference from a regular option is payoff shape: a standard call payoff keeps growing with spot above strike, but a binary pays a fixed amount if it settles on the winning side. For this reason, binary moneyness is mostly a measure of distance from the settlement boundary and likely sensitivity to small underlying moves.

Plot logic issue identified

The user noticed that the BTC Futures Distance from Contract Threshold plot repeatedly showed an asymptotic-looking collapse near close. The assistant reasoned that true moneyness, computed as underlying price minus threshold or as ((underlying - threshold) / threshold) * 10000, should not naturally have asymptotic behavior. Therefore the issue was likely with the plot logic or with the precomputed z_bps field rather than with binary moneyness itself.

Suggested diagnostic

The assistant recommended inspecting contract_evolution_1s.parquet for a clean threshold/strike column, such as threshold, strike, strike_price, contract_threshold, floor, cap, or yes_threshold. If present, the app should recompute threshold distance directly inside app.py instead of relying on the suspicious z_bps column.

```
python -c "import pandas as pd, pathlib; p=pathlib.Path('data/app/weeks');
f=list(p.glob('*/contract_evolution_1s.parquet'))[-1]; df=pd.read_parquet(f);
cols=df.columns.tolist(); print('checking:', f);
print([c for c in cols if any(x in c.lower() for x in
['threshold','strike','floor','cap','bound','target'])])"
```

Outcome

The conceptual interpretation was separated from the implementation bug. The chart was useful in principle, but the assistant warned that z_bps should not be trusted as the main explanation until its construction was verified or replaced with a direct threshold-distance formula.